CHAPTER IV

ANALYSIS AND INTERPRETATION OF DATA

This chapter presents the analysis of the collected raw data and aims to provide a meaningful understanding of the research findings. Analysis and interpretation are two essential stages in the research process. Data analysis is considered the core of a research report, as it involves examining the organized and tabulated data to identify underlying facts and meanings (Best, 2006).

The final and most important stage of analysis is interpretation. Interpretation refers to explaining the findings, addressing “why” questions, assigning significance to specific results, and organizing observed patterns within an analytical framework (Patton, 1990). Interpretation is not a simple mechanical procedure; rather, it requires careful, logical, and critical evaluation of analytical results while considering the limitations of the study (Burns, 2000).

The primary objective of the present study was to examine mental resilience among football referees at professional and non-professional matches. The statistical techniques employed included arithmetic mean, standard deviation, and independent samples t-test. The hypothesis was tested at the 0.05 level of significance.

RESULTS OF THE STUDY

This section presents the conclusions drawn after comprehensive statistical analysis. When the obtained p-value was less than 0.05 ($p < .05$), the hypothesis was accepted; otherwise, it was rejected.

Table 4.1

Descriptive statistics of mental resilience variables among professional and non-professional football referees

Variable	Group	N	Mean	SD
SMTQ	Professional	30	36.27	4.44
SMTQ	Non-professional	30	35.70	3.25

NMRQ	Professional	30	41.80	10.52
NMRQ	Non-professional	30	34.47	10.14
SAS	Professional	30	30.10	10.10
SAS	Non-professional	30	34.20	8.06

Interpretation

Table 4.1 reveals that the mean SMTQ score of professional referees (M = 36.27, SD = 4.44) was slightly higher than that of non-professional referees (M = 35.70, SD = 3.25).

For NMRQ, professional referees (M = 41.80, SD = 10.52) showed higher resilience than non-professional referees (M = 34.47, SD = 10.14).

Regarding SAS, non-professional referees (M = 34.20, SD = 8.06) scored higher than professional referees (M = 30.10, SD = 10.10).

Table 4.2

Independent samples t-test comparing professional and non-professional referees on SMTQ

Variable	t	df	p	Mean Difference
SMTQ	0.564	58	.575	0.567

Interpretation

Table 4.2 shows that the calculated t-value for SMTQ was t(58) = 0.564, with p = .575. Since the obtained p-value is greater than the significance level of .05, there is no significant difference between professional and non-professional football referees in SMTQ scores.

Table 4.3

Independent samples t-test comparing professional and non-professional referees on NMRQ

Variable	t	df	p	Mean Difference
NMRQ	2.749	58	.008	7.333

Interpretation

Table 4.3 indicates that the calculated t-value for NMRQ was $t(58) = 2.749$, with $p = .008$. Since the obtained p-value is less than .05, there is a statistically significant difference between professional and non-professional referees in NMRQ scores. Professional referees demonstrated higher mental resilience compared to non-professional referees.

Table 4.4

Independent samples t-test comparing professional and non-professional referees on SAS

Variable	t	df	p	Mean Difference
SAS	−1.738	58	.088	−4.100

Interpretation

Table 4.4 shows that the calculated t-value for SAS was $t(58) = -1.738$, with $p = .088$. Since the p-value is greater than .05, there is no significant difference between professional and non-professional referees in SAS scores.

4.2 DISCUSSION OF FINDINGS

The present study examined mental resilience among professional and non-professional football referees using SMTQ, NMRQ, and SAS scales.

The descriptive statistics indicated that professional referees showed slightly higher scores in SMTQ and clearly higher scores in NMRQ compared to non-professional referees. However, non-professional referees demonstrated somewhat higher SAS scores.

The independent samples t-test revealed that:

No significant difference existed in SMTQ

A significant difference existed in NMRQ

No significant difference existed in SAS

These findings suggest that professional refereeing experience may particularly enhance resilience aspects measured by NMRQ, while other components of mental toughness remain comparable between groups.

4.3 TESTING OF HYPOTHESES

H1: There will be a significant difference in mental resilience between professional and non-professional football referees.

SMTQ: $t(58) = 0.564$, $p = .575$ → Not significant → Rejected

NMRQ: $t(58) = 2.749$, $p = .008$ → Significant → Accepted

SAS: $t(58) = -1.738$, $p = .088$ → Not significant → Rejected